

Optimization & Machine Learning M

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Context and Objective

You have recently joined the Data Operations team at *Capital Bikes*, a public bicycle sharing system in Washington D.C. The company is facing a major logistical challenge: customer dissatisfaction due to empty stations (unable to rent a bike) and completely full stations (unable to return a bike).

Your manager has entrusted you with a two-part mission:

- **Machine Learning (Prediction):** Develop a model to predict the hourly demand for bike rentals based on temporal and meteorological factors.
- **Operations Research (Optimization) [Bonus]:** Use these predictions to optimize the overnight redistribution of bikes using logistics trucks.

Deliverables: A clean, commented, and well-structured Jupyter Notebook (`.ipynb`) containing your code and textual analyses.

Machine Learning Tasks

1. Data Exploration and Preprocessing (40%)

In the directory `data`, you are provided with two files: the dataset `bike_sharing_dataset.csv` and a text file `about_bike_sharing_dataset.txt` that describes the dataset's characteristics. Note that the target variable you need to predict is `cnt`.

The raw dataset may contain imperfections and anomalies. You must clean and prepare the dataset for model training.

- (a) **Exploration:** Load the dataset. Study the dimensions of the dataset, understand the meaning of each characteristics.
- (b) **Missing Values:** Identify and handle missing values logically.
- (c) **Anomalies:** Identify physically impossible values and outliers. Correct, impute or delete them appropriately. *Hint: Look at the characteristics' statistical distributions.*
- (d) **Feature Selection:** Identify and remove redundant columns, collinearities, and data leakage variables. Justify your choices. *Hint: You can generate a correlation matrix.*
- (e) **Feature Engineering:** Encode categorical variables appropriately. *Hint: Apply mathematical transformations (e.g., sine/cosine) to cyclic variables like the hour of the day.*
- (f) **Analysis:** Visualize the distribution of the target variable (`cnt`) and its relationship with temporal (hour, day, month) and meteorological variables. Comment.

2. Model Development and Evaluation (40%)

- (a) **Data Splitting:** Split the dataset into training (80%) and testing (20%) subsets.
- (b) **Baseline Model:** Train a baseline linear model (e.g., Linear Regression). Evaluate its performance using suitable metrics (e.g., RMSE, MAE, R-squared).
- (c) **Advanced Modeling:** Choose and train an appropriate non-linear machine learning algorithm (e.g., Random Forest).
- (d) **Fine-tuning:** Optimize your best model's hyperparameters using Grid Search to improve performance.

3. Prediction and Interpretation (20%)

- (a) **Evaluation:** Use the trained model to make predictions on the testing data and assess its final performance.
 - (b) **Feature Importance:** Extract and visualize the most important features of your model. Do these align with your initial exploratory data analysis?
 - (c) **Conclusion:** Summarize your findings and provide business recommendations.
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Operations Research Task

4. Stock Optimization (Bonus)

Your predictions are meant to drive business decisions. Assume you isolate a larger sub-network of 20 stations. Comparing current stocks to the predicted demand for the next morning, you identify 8 stations with a surplus (supply) and 12 stations with a deficit (demand).

You have been provided with an `or_data.json` file containing:

- a list of 20 stations;
- their current `supply_demand` status (positive integers for surplus, negative integers for deficit);
- a 20×20 symmetrical `distance_matrix` representing the travel costs in kilometers between all stations.

Let $\mathcal{S}$ be the set of surplus stations, and $\mathcal{D}$ be the set of deficit stations. Let c_{ij} be the transportation cost (distance in km) between station $i \in \mathcal{S}$ and station $j \in \mathcal{D}$. Let X_{ij} be the number of bikes to move from i to j .

- (a) **Mathematical Formulation:** Write the Linear Program to minimize the total redistribution cost, ensuring you do not move more bikes than the available surplus, while satisfying the deficits as much as possible.
- (b) **Implementation:** Import the JSON file and use the PuLP library in Python to solve this problem. PuLP is highly recommended because its intuitive, algebraic syntax makes modeling objective constraints straightforward (see the [official case studies](#) for guidance). Provide the optimal transportation plan (the non-zero values of X_{ij}) and the total optimized cost.